MIL-S-37807A
22 June 1978
SUPERSEDING
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16 June 1977

MILITARY SPECIFICATION

SULFUR, SALICYLIC ACID AND POLYETHYLENE CREAM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Sulfur, Salicylic Acid and Polyethylene Cream (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (ID Form 1426) appearing at the end of this document, or by letter.

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be an unscented, abradent acne scrub of the following formula:

Sulfur, USP	2 percent
Salicylic Acid, USP	1.5 percent
Granular Polyethylene	26 percent
Suitable base 1/	q.s.

1/ The suitable base shall be a penetrating type with surface active combinations of soapless cleansers and wetting agents.

3.2 Assay of finished cream.

3.2.1 Sulfur. Shall assay to contain not less than 1.80 percent and not more than 2.20 percent of sulfur when determined as specified in 4.7.1.

3.2.2 Salicylic Acid. Shall assay to contain not less than 1.35 percent and not more than 1.65 percent of salicylic acid when determined as specified in 4.7.2.

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3.2.3 Granular polyethylene. Shall assay to contain not less than 23.4 percent and not more than 28.6 percent of granular polyethylene when tested as specified in 4.7.3.

3.3 pH. The pH of the finished cream shall be not less than 2.4 percent and not more than 3.4 percent when tested as specified in 4.7.4.

3.4 Loss on drying. The loss on drying of the finished cream shall be not less than 43.0 percent and not more than 48.0 percent when determined as specified in 4.7.5.

3.5 Lather volume and permanence tests. The total volume of lather produced by the finished cream shall be not less than 50 ml when read immediately after testing and not less than 35 ml after standing for 30 minutes, when tested as specified in 4.7.6.

3.6 Particle size of granular polyethylene. **Not more than 5 percent of granular polyethylene shall be retained on a No. 50 sieve when tested as specified in 4.7.7. The test shall be performed on the raw material entering into the finished product.**

3.7 Ingredients.

3.7.1 Sulfur. The sulfur used in the manufacture of the cream shall be precipitated sulfur and shall comply with the tests, standards and requirements of the USP, including any supplements or revisions thereto.

3.7.2 Salicylic acid. The salicylic acid used in the manufacture of the cream shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.8 Other ingredients. All other ingredients used in the manufacture of the cream shall be of USP or NF quality or, if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade.

3.9 Delivery. Not more than four (4) months shall have elapsed from the date of manufacture of the cream to the date of delivery to the Government.

3.10 Workmanship. The cream and its container shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any othe facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of cream produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Filled immediate container

4.3.1 Raw material and end item testing. Noncompliance with any of the following is cause for rejection.

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TABLE II. Criteria for raw material and end item testing.

Characteristic	Requirement paragraph	Inspection level	Minimum sample
Assay for sulfur	3.2.1	1/ S-2 To obtain a composite sample to be utilized in performing these tests.	5 boxes
Assay for salicylic acid	3.2.2		
Assay for granular polyethylene	3.2.3		
pH	3.3		
Loss on drying	3.4		
Lather volume and permanence test	3.5		
Particle size of granular polyethylene	3.6	S-2	2/

1/ Where possible, the sample shall be used for two or more tests.

2/ Sufficient quantities of granular polyethylene shall be selected from each lot to perform tests indicated in applicable test methods. Failure of any sample to comply with test requirements shall be cause for rejection of the lot represented.

4.4 Examination. The finished cream shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3 and 4.3.1.

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4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

TABLE III. Classification of defects

Categories	Defects 1/
Major	
101	Cream not free from foreign matter.
102	Cream not homogeneous; not free from separation.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks or tears.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Packaging inspection. The inspection of the immediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Tests.

4.7.1 Assay for sulfur in the finished cream.

Standard solution

Prepare a solution of Sulfur, USP in Iso-octane having a known final concentration of approximately 0.2 mg per ml.

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Procedure

Accurately weigh approximately 100 mg of cream into a beaker, disperse in 20 ml of purified water, and transfer quantitatively to a 60 ml separator using small portions of purified water. Pipet 10.0 ml methylene chloride, reagent grade to the separator, stopper and shake well. Permit the layers to separate, and run off approximately 5 ml of the lower layer (organic solvent) through phase separating paper such as Whatman 1PS, or equivalent. Collect the clear filtrate in an Erlenmeyer flask and stopper to prevent evaporation.

Using a prepared thin layer chromatography plate containing Silica Gel GF, with a uniform layer thickness of approximately 250 microns, separately spot 10 µl of standard solution and 10 µl of clear filtrate sample solution approximately 2.5 cm from the bottom of the plate. Develop to a height of approximately 10 cm using iso-Octane as the mobile solvent in a solvent chamber previously saturated with iso-Octane vapor.

Permit the plates to dry thoroughly in air at room temperature, and quantitate the spots corresponding to Sulfur (Rf approximately 0.7) using a recording densitometer, with exciting light at 254 nm and emitted light measured at 520 nm. Set the instrument for optimal operating conditions. Integrate the areas under the curve for standard and sample.

Calculation

$$\frac{Qu \times Cs \times 1000}{Qs \times Sw\ (mg)} = \text{percent Sulfur in cream}$$

Where Qu - Quantitation value sample
Qs - Quantitation value standard
Cs - Concentration standard mg per ml
Sw - Sample weight

4.7.2 Assay for salicylic acid in the finished cream.

Reagent

Dissolve 1.0 gram of $Fe(NO_3)_3$ and 1.0 ml concentrated nitric acid in 100.0 ml purified water.

Standard solution

Accurately weigh approximately 40 mg of salicylic acid USP reference standard into a 100 ml volumetric flask, dissolve in, and dilute to volume with SDA 3A alcohol.

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Procedure.

Weigh accurately approximately 200 mg of product into a tared 100 ml volumetric flask. Add 20 ml of SDA 3A alcohol. Shake well to disperse.

It may be necessary to use a suitable ultrasonic device.

Add about 50 mls purified water, 10.0 ml $Fe(NO_3)_3$ reagent and dilute to volume with purified water. Shake well.

Pipet 10.0 ml of salicylic acid standard solution in a separate 100 ml flask, add about 50 ml purified water, 10.0 ml $Fe(NO_3)_3$ reagent and make to volume with purified water. Shake well.

Filter the purple solutions through double No. 42 filter paper until clear.

Determine the absorbance at 525 nm against a water blank.

Calculation

$$\frac{Au \times Cs \times 10}{As \times Sw} = \text{percent salicylic acid in product}$$

where Au = absorbance of unknown
As = absorbance of standard
Cs = standard weight in mg/100 ml
Sw = sample weight in mg

4.7.3 Assay for granular polyethylene.

Weigh accurately into a tared 100 ml beaker about 1 gram of sample followed by 20 ml SDA 3A alcohol and 20 ml $CHCl_3$. Let stand about 5 minutes with occasional stirring until no aggregate particles are visible and transfer to a suction filtration apparatus. Quantitatively wash beaker and contents with purified water into a preweighed coarse filtering crucible using vacuum as required. Turn off the vacuum and add 10 ml warm to hot sulfuric acid to the residue in the crucible, stir with a glass rod for about one minute, turn on vacuum, flush with purified water and repeat with second portions of sulfuric acid and purified water. Flush again with methanol; transfer to a drying oven at 60°C to 80°C for at least 10 minutes, cool and reweigh.

The percent polyethylene is calculated as follows:

$$\frac{\text{Wt Residue (gram)} \times 100}{\text{Sample Wt (gram)}} = \text{percent polyethylene}$$

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4.7.4 pH. Weigh approximately 5 gram of sample into a tared beaker. Add 20 gram of purified water, and stir vigorously until the sample is well suspended in the aqueous phase, and determine the pH of the resulting solution potentiometrically using the USP method.

4.7.5 Loss on drying.

Weigh accurately an aluminum weighing dish and record the weight (t). Weigh accurately into the dish about 3 to 4 grams of a well-mixed sample and record the total weight (Wt_1). Place the dish into a 110°C ±5°C oven for two hours. Remove the dish from the oven and let cool for five minutes. Reweigh the dish and record the weight (Wt_2).

$$\frac{(Wt_1 - Wt_2) \times 100}{(Wt_1 - t)} = \text{percent drying loss}$$

Wt_1 = initial weight of dish and sample

Wt_2 = final weight of dish and sample

t = weight of dish

4.7.6 Lather volume and permanence. Transfer 60 mg of cream to a 100 ml glass stoppered graduated cylinder and add 20 ml of deionized water at room temperature (25°C ± 1°C). Shake vigorously for 30 seconds. The total volume of water and lather shall be read immediately and after 30 minutes.

4.7.7 Particle size of granular polyethylene.

Procedure

Place 10.0 gram of material on a No. 50 sieve, which has been previously weighed to the nearest 0.1 gram. Shake the sieve until no more material passes through the screen. Reweigh the sieve.

Calculations

$$\frac{(\text{Final wgt. of sieve} - \text{Initial wgt. of sieve}) \times 100}{10.0 \text{ gram}} = \text{percent retained on sieve}$$

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One box containing twelve (12) 2 oz tubes, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available tube considered acceptable by the Food and Drug Administration [illegible] for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. Each unit shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Twelve (12) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Twelve (12) units of issue shall be packed in a fiberboard box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class domestic.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder, and shall include the information required below:

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5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the date of manufacture prefixed by "MFD".

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number need not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.

5.4.2 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186 except that date of manufacture shall be shown in lieu of date packed.

6. NOTES

6.1 Ordering date. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-01-035-1719	SULFUR, SALICYLIC ACID AND POLYETHYLENE CREAM, 2 oz (56.7 Gram), 12s

6.3 Changes from previous issue. The margins of this specification are marked with an asterick to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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